

Amr Abujabal

Computer Science | Cloud & Backend | Computer Vision
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EDUCATION

University of British Columbia (Okanagan)
Bachelor of Science in Computer Science

Expected May 2027

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL, R, JavaScript, HTML, CSS

Backend & Web: FastAPI, Next.js, React, PostgreSQL, MySQL, Redis, Celery, REST APIs, JWT auth

Cloud & DevOps: Google Cloud Platform, AWS S3, Docker, Git, GitHub Actions, CI/CD, Linux/Unix

ML & Data: PyTorch, NumPy, pandas, scikit-learn, XGBoost, YOLOv10, SAM 2, TransReID/OSNet, vision-language models, multi-object tracking

Tools: Anaconda, Jupyter, RStudio

PROJECTS

PitchVision / *Computer Vision / ML · Cloud*

2026 – Present

- Building a single-camera player identification and performance analytics system for amateur and semi-pro soccer clubs, turning raw touchline footage into per-player metrics.
- Engineered a tracking and re-identification pipeline (YOLOv10 detection, SAM 2 segmentation, TransReID/OSNet re-ID) to maintain player identity across full-match clips.
- Serving coach-facing outputs — movement stats (distance, speed zones), heatmaps, pressing and formation views — over a FastAPI REST API into a Next.js dashboard, backed by PostgreSQL and Docker; 126 tests passing in CI.

FitCheckAI / *AI Fashion Assistant · Full-Stack*

2026 – Present

- Built an Instagram bot that turns an outfit photo into shoppable product matches: a DM webhook hands off to Celery workers that run a vision model to break the outfit into garments, then search each one across three price tiers and reply in a stylist voice.
- Designed the pipeline async end to end on FastAPI over PostgreSQL, Redis and AWS S3, with an SSRF-guarded image download path and per-user rate limiting.
- 106 tests green end to end on Docker Compose, with a pluggable vision provider (OpenAI Vision primary, Claude as fallback).

Reproducing ResNet — Residual Learning on CIFAR / *Deep Learning · Paper Reproduction*

2026

- Rebuilt the CIFAR-10 experiments from He et al. (2015) using only the paper — model, data pipeline and training loop — test-first (TDD), with tests asserting exact layer counts and parameter budgets (0.27M at ResNet-20, 0.85M at ResNet-56) before any training step.
- Reproduced ResNet-20 at $8.39 \pm 0.31\%$ and ResNet-56 at $7.45 \pm 0.69\%$ error over 3 seeds, both inside the $\pm 0.5\%$ tolerance fixed before the first run.
- Ran a controlled 2x2 ablation (original vs. pre-activation ResNet-56 x CIFAR-10 vs. CIFAR-100) and found no measurable difference at depth 56 — the gaps are smaller than the seed noise and point in opposite directions — reported as a null result. 15 runs, ~26 GPU-hours; write-up published at resnet.amrabujabal.com.

World Cup 2026 Match Outcome Predictor / *Full-Stack ML*

2026

- Trained a 3-class match outcome classifier (Win / Draw / Loss) with XGBoost on 49,000+ historical matches using ELO-based engineered features, pricing every knockout tie before kickoff.
- Shipped it as a full-stack app — FastAPI, PostgreSQL and React behind an interactive bracket with user predictions and a leaderboard — deployed on Vercel, Render and Neon, with a scheduled sync that pulled real results from football-data.org and advanced the bracket as each match finished.
- Ran live through the 2026 FIFA World Cup and finished 26 of 31 knockout calls correct (83.9%), Brier 0.3665, perfect through the quarter-finals and semis; finished bracket live at worldcup.amrabujabal.com.